School of Computer Science Engineering and Technology

Course- **BTech**
Course Code- **CSET340**

Year- **2026**
Date- **16-03-2026 to 20-03-2026**

Type- **Specialization Elective**
Course Name-**Advanced Computer vision and Video Analytics**
Semester- **EVEN**
Batch- **2023-2027 (V Sem)**

Lab Assignment No. 08

Exp. No.	Name	CO-1	CO-2	CO-3
1.	To perform various Image Compression and Watermarking on given set of images	✓	✓	✓

Objective: Apply various image compression techniques ranging from lossy to lossless compression and digital watermarking tasks on different types of images via the use of Python using OpenCV/Pillow/Ski Images Library? Perform the provided task as follows.

Data Set: Download the Zip_Folder containing "Test_Images_lab_08" images with some colored, gray scaled and black and white images.

1. Use the grayscale image & apply the following Loss-less Compression Techniques to the image **(CameraMan.jpeg)**. Also display the **original image and Compressed image** obtained after applying each compression technique. Calculate Compression Loss: Use metrics like Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR) to quantify the loss.
    a. Run Length encoding
    b. Huffman Coding **3 Marks**
2. Use the grayscale image & apply the following Lossy Compression Techniques to the image **(CameraMan.jpeg)**. Also display the **original image and Compressed image** obtained after applying each compression technique. Calculate Compression Loss: Use metrics like Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR) to quantify the loss.
    a. Discrete cosines transform without Quantization
    b. DCT with Zig Zag Coding and Quantization **3 Marks**
3. Use **(Face.jpg)** image to perform step-by-step lossy compression using a like **JPEG** and compare the size of original vs lossy image**.** Use **(Butterfly.png)** image to perform image compression using a lossless method like **PNG** with different compression levels and compare the size of the original vs the output image**.** **1 Marks**
4. Use the **(penguin.jpg)** image and insert **(BUlogo.png)** image as **visible and invisible watermark** separately in it. Now extract the visible and invisible watermarks from the image by using **DCT** to get back the original image. **3 Marks**

Note:- Suggested Platform: Python: Jupyter Notebook/Visual Studio Code/Google Colab.
Mode of Delivery: Face-to-face: Instructor-led discussion and live coding demonstration. Hands-on Practice via Google Colab/VS code/Notebook.
Submission: On LMS within the prescribed time frame, else marks will.